

A Curvature Identity for the Fibonacci Digital-Root Cycle Modulo 9

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Abstract

For each seed we form the 24-periodic orbit of the Fibonacci recurrence modulo 9, write its terms with the digital-root convention $0 \mapsto 9$, and measure its *digital curvature* $S = \sum_n \kappa_n^2$, the sum of squared discrete second differences. We make two points. First, on the 72 full-period seeds modulo 9 the curvature takes exactly three values, $\{648, 972, 1296\}$, each on 24 seeds; the maximum 1296 is attained precisely by the classical Fibonacci cycle $(1, 1)$ and its rotations. This is a finite statement and we prove it by exhaustive enumeration. Second, and more usefully, S is the squared norm of a circulant operator and therefore has an exact Fourier form, $S = \frac{16}{N} \sum_k \sin^4(\pi k/N) |\hat{a}_k|^2$. This identity explains the classification: the curvature is a high-pass energy, and the Fibonacci orbit maximizes it because it is the most high-frequency of the three orbit classes modulo 9, carrying about 98% of its curvature energy in the near-Nyquist half of the spectrum. We use the same lens to describe how far the maximality property extends to other moduli (it is a small-modulus phenomenon, not a law) and to retire a conjecture from earlier versions of this note. This version corrects errors in Versions 1–3; see the revision history in Section 8.

Keywords: Fibonacci numbers; Pisano period; digital root; discrete curvature; circulant operator; discrete Fourier transform.

1 Introduction

Let $F(n)$ be the Fibonacci numbers, $F(0) = 0$, $F(1) = 1$. The sequence $F(n) \bmod 9$ has Pisano period 24 [1]. Replacing each residue 0 by 9 (the digital-root convention used in base 10, since the digital root of an integer equals its value modulo 9 with $0 \mapsto 9$) yields the familiar cycle

$$1, 1, 2, 3, 5, 8, 4, 3, 7, 1, 8, 9, 8, 8, 7, 6, 4, 1, 5, 6, 2, 8, 1, 9.$$

This 24-tuple is well known, but its higher-order difference statistics are not usually examined. We attach to every modulo-9 orbit a single number—its *digital curvature*, the energy of its discrete second difference—and ask what values that number takes.

The answer modulo 9 is a clean trichotomy (Theorem 1). The point of this version of the note, however, is not the trichotomy itself but the reason behind it. The curvature functional is the squared norm of the cyclic second-difference operator, which is circulant and hence diagonalized by the discrete Fourier transform. This gives an exact spectral formula for S (Proposition 1) from which the classification, the identity of the maximizer, and the limits of that maximality all follow transparently. The spectral identity is a standard fact of harmonic analysis on the cycle; what is (to our knowledge) not previously recorded is its application to the digital-root orbits of a Pisano period, and the resulting fingerprint of the classical Fibonacci cycle.

We have tried to be careful about what is and is not established. The trichotomy is a theorem proved by finite computation. The spectral identity is a theorem. The statement that the Fibonacci orbit maximizes curvature is true for small moduli but provably fails for larger ones, and we report it as the bounded empirical fact that it is. Earlier versions of this note advanced a closed-form conjecture and a symmetry argument that turn out to be, respectively, a numerical coincidence and an error; both are corrected below.

2 Definitions

Fix a modulus m (we take $m = 9$ unless stated otherwise) and let $P = \pi(m)$ be its Pisano period. For a seed $(a_0, a_1) \in (\mathbb{Z}/m\mathbb{Z})^2$ generate the orbit

$$a_{n+1} = a_n + a_{n-1} \pmod{m}.$$

We restrict attention to seeds whose orbit has *minimal period exactly* P (“full-period” seeds); these are the orbits that realize the full Pisano cycle. The *rotation class* of such a seed is the set of P pairs (a_n, a_{n+1}) occurring along its orbit—equivalently, its P cyclic shifts. Display each residue with

$$d(r) = \begin{cases} m & r \equiv 0, \\ r & r \in \{1, \dots, m-1\}, \end{cases}$$

so displayed values lie in $\{1, \dots, m\}$, and set

$$\kappa_n = d(a_{n+1}) - 2d(a_n) + d(a_{n-1}) \in \mathbb{Z}, \quad S = \sum_{n=0}^{P-1} \kappa_n^2,$$

all indices taken cyclically modulo P . We call S the *digital curvature* of the orbit. Two remarks are worth making at the outset.

Remark 1 (Convention dependence). S depends on the display convention $0 \mapsto m$. With residues taken in $\{0, \dots, m-1\}$ instead, the same orbit yields different second differences and a different S . The quantity studied here is therefore an invariant of the *digital-root presentation* of the orbit, not of the abstract residue sequence. This is deliberate—the digital-root cycle is the object of interest—but it should be stated plainly.

Remark 2 (Curvature is a cyclic energy). $\kappa_n = (Dd)_n$ where D is the cyclic second-difference operator, so $S = \|Dd\|^2$ is the discrete Dirichlet-type energy of the displayed orbit on the cycle C_P . This observation drives everything in Section 4.

3 The Trichotomy Modulo 9

Theorem 1. *Among the 72 full-period seeds modulo 9, the digital curvature takes exactly the three values*

$$S \in \{648, 972, 1296\},$$

each attained by exactly 24 seeds (equivalently, by one full-period orbit apiece). Moreover $S = 1296$ if and only if the seed lies in the rotation class of $(1, 1)$, and then

$$S = 1296 = \sum_{n=0}^{23} \kappa_n^2.$$

Proof. There are 81 ordered seeds $(a_0, a_1) \in \{1, \dots, 9\}^2$; of these, 72 generate orbits of minimal period 24 and 9 do not. Evaluating S on each full-period orbit is a finite computation, carried out in the script of Section 7; it returns the multiset $\{648:24, 972:24, 1296:24\}$, with the value 1296 occurring exactly on the 24 rotations of the $(1, 1)$ orbit. A finite exhaustive check is a complete proof of a statement quantified over a finite set. $\square$

We stress that the enumeration *is* the proof. Earlier versions of this note tried to derive the three values from a congruence argument; that argument is vacuous (all three values are divisible by 9, so reduction modulo 9 cannot distinguish them) and has been removed. What survives from it is a correct structural observation about *which* orbits carry the three values.

Remark 3 (The three values are the three unit classes). The unit group $(\mathbb{Z}/9\mathbb{Z})^\times = \{1, 2, 4, 5, 7, 8\}$ acts on orbits by $a_n \mapsto g a_n$. Because $F_{n+12} \equiv -F_n \pmod{9}$ *at the level of residues*, negation $g = -1 \equiv 8$ maps the $(1, 1)$ orbit to its own shift by 12; hence the subgroup $\{\pm 1\}$ fixes each orbit and the six units collapse to the three classes

$$\{\pm 1\} = \{1, 8\}, \quad \{\pm 2\} = \{2, 7\}, \quad \{\pm 4\} = \{4, 5\},$$

i.e. $(\mathbb{Z}/9\mathbb{Z})^\times / \{\pm 1\}$. Scaling the $(1, 1)$ orbit by a representative of each class gives precisely the three curvature values:

$$g \in \{1, 8\} \rightarrow 1296, \quad g \in \{2, 7\} \rightarrow 972, \quad g \in \{4, 5\} \rightarrow 648,$$

verified directly. So the trichotomy is the statement that curvature is an invariant of $(\mathbb{Z}/9\mathbb{Z})^\times / \{\pm 1\}$, taking distinct values on its three classes. (This residue-level symmetry is what makes the unit-class action well defined; it is only the antisymmetry of the displayed *second differences* that the $0 \mapsto 9$ convention breaks, as Section 4.1 shows.)

Table 1 records the second differences of the maximal $(1, 1)$ orbit, for use below. One checks directly that $\sum_n \kappa_n = 0$ and $\sum_n \kappa_n^2 = 1296$.

4 Curvature as Spectral Energy

The cyclic second-difference operator D on length- N sequences is circulant, hence diagonalized by the discrete Fourier transform. This yields an exact formula for S .

n	0	1	2	3	4	5	6	7	8	9	10	11
a_n	1	1	2	3	5	8	4	3	7	1	8	9
κ_n	8	1	0	1	1	-7	3	5	-10	13	-6	-2

n	12	13	14	15	16	17	18	19	20	21	22	23
a_n	8	8	7	6	4	1	5	6	2	8	1	9
κ_n	1	-1	0	-1	-1	7	-3	-5	10	-13	15	-16

Table 1: Second differences for the $(1, 1)$ orbit. $\sum \kappa_n = 0$, $\sum \kappa_n^2 = 1296$.

Proposition 1. Let $d = (d_0, \dots, d_{N-1})$ be the displayed orbit (of length $N = P$), with unnormalized DFT $\hat{d}_k = \sum_n d_n e^{-2\pi i k n / N}$. Then

$$S = \frac{1}{N} \sum_{k=0}^{N-1} \mu_k^2 |\hat{d}_k|^2, \quad \mu_k = 2 \cos \frac{2\pi k}{N} - 2 = -4 \sin^2 \frac{\pi k}{N},$$

equivalently

$$S = \frac{16}{N} \sum_{k=0}^{N-1} \sin^4 \left(\frac{\pi k}{N} \right) |\hat{d}_k|^2.$$

Proof. $\kappa_n = (Dd)_n = d_{n+1} - 2d_n + d_{n-1}$ with indices modulo N , so D is the circulant matrix with symbol $e^{2\pi i k / N} - 2 + e^{-2\pi i k / N} = 2 \cos(2\pi k / N) - 2 = \mu_k$. The Fourier modes $e^{2\pi i k n / N}$ are its eigenvectors with eigenvalues μ_k , so $\widehat{Dd}_k = \mu_k \hat{d}_k$. By Parseval, $S = \|Dd\|^2 = \frac{1}{N} \sum_k |\mu_k \hat{d}_k|^2 = \frac{1}{N} \sum_k \mu_k^2 |\hat{d}_k|^2$, and $\mu_k^2 = 16 \sin^4(\pi k / N)$. $\square$

This is standard harmonic analysis on the cycle; we record it only because it converts the opaque integer S into something interpretable. The weight $16 \sin^4(\pi k / N)$ vanishes at the constant mode $k = 0$, grows monotonically across $k = 1, \dots, \lfloor N/2 \rfloor$, and is maximal at the Nyquist mode $k = N/2$. Thus *digital curvature is a high-pass energy*: it ignores the orbit's mean and slow variation and rewards rapid, near-alternating oscillation. Call the modes with $\sin^2(\pi k / N) \geq \frac{1}{2}$ —equivalently $N/4 \leq k \leq 3N/4$ —the *near-Nyquist half* of the spectrum.

For the maximal $(1, 1)$ orbit ($N = 24$, $S = 1296$), Table 2 gives the per-mode contribution $c_k = \frac{1}{N} \mu_k^2 |\hat{d}_k|^2$. Conjugate modes k and $N - k$ contribute equally, so $S = c_0 + 2 \sum_{k=1}^{11} c_k + c_{12}$. The near-Nyquist half carries about 97.6% of S for this orbit; the constant and low-frequency modes carry the remaining 2.4%.

Remark 4 (The maximizer is the highest-frequency class). The three modulo-9 classes are ordered by how much of their curvature sits in the near-Nyquist half: it is 97.6% of S for the 1296 class, against 94.0% and 86.6% for the 972 and 648 classes. So among the three full-period orbit classes modulo 9, the Fibonacci class is the one whose curvature is most concentrated near the Nyquist frequency. This is a diagnosis, not a derivation: because S is the high-pass energy, the statement reformulates maximality spectrally but does not explain why the $(1, 1)$ seed produces the most near-Nyquist spectrum in the first place (see Section 9).

k	$16 \sin^4(\pi k/24)$	$ \hat{d}_k ^2/N$	c_k
0	0.0000	570.375	0.000
1	0.0046	17.919	0.083
2	0.0718	3.375	0.242
3	0.3431	19.125	6.563
4	1.0000	3.375	3.375
5	2.1974	2.331	5.122
6	4.0000	3.375	13.500
7	6.3385	2.331	14.774
8	9.0000	3.375	30.375
9	11.6569	19.125	222.937
10	13.9282	3.375	47.008
11	15.4595	17.919	277.022
12	16.0000	3.375	54.000

Table 2: Spectral decomposition of the $(1, 1)$ orbit’s curvature. Conjugate modes k and $24 - k$ contribute equally, so $S = c_0 + 2 \sum_{k=1}^{11} c_k + c_{12} = 1296$. The contribution rises with frequency; the dominant pairs are $k = 11, 13$ (277.0 each) and $k = 9, 15$ (222.9 each), with the Nyquist mode $k = 12$ contributing 54.0. The near-Nyquist half ($\sin^2(\pi k/24) \geq \frac{1}{2}$, i.e. $6 \leq k \leq 18$) accounts for about 97.6% of S .

4.1 The half-period asymmetry, corrected

The identity $F_{n+12} \equiv -F_n \pmod{9}$ (from $24 = 2 \cdot 12$ [1]) suggests the second differences should satisfy $\kappa_{n+12} = -\kappa_n$, which would force the two half-period curvature sums to be equal. *This is false*, and an earlier version of this note relied on it in error. The relation $\kappa_{n+12} = -\kappa_n$ holds for $n = 1, \dots, 9$ but breaks at $n = 0, 10, 11$ —exactly the indices adjacent to the displayed zeros at positions 11 and 23, where $d(0) = 9$ severs the antisymmetry. Consequently the halves are *unequal*:

$$\sum_{n=0}^{11} \kappa_n^2 = 459, \quad \sum_{n=12}^{23} \kappa_n^2 = 837, \quad 459 + 837 = 1296,$$

and the entire imbalance $837 - 459 = 378$ comes from those three zero-crossing index pairs. The factorization $1296 = 2 \cdot 648$ is therefore a coincidence, not a consequence of half-period symmetry; the curvature does not split evenly across the period.

5 How Far Maximality Extends

It is natural to ask whether the Fibonacci seed maximizes digital curvature for every modulus. It does not. Define, for any modulus m with even Pisano period, $S_{\max}(m)$ as the maximum of S over full-period orbits, computed with the same display convention.

Small moduli. Among moduli $m \leq 14$ with even Pisano period, $(1, 1)$ attains $S_{\max}(m)$ in every case; the first failure is $m = 15$.

Erosion with scale. Over the 118 moduli $m \leq 120$ with even Pisano period, $(1, 1)$ attains $S_{\max}(m)$ for 78 of them and fails for 40. The first three failures are $m = 15, 33, 40$ (and these are the only failures below 41); the full failure list continues

45, 49, 52, 53, 56, 59, 60, 64, 65, 67, 68, 71, 74, 80, 81, 83, 85, 86, 88, 90, $\dots$, 120.

The proportion of failures broadly increases with m (reaching 60–70% among moduli in the 80s–100s), though not monotonically.

No arithmetic predictor. No simple arithmetic rule we tested separates the failures from the successes. In particular “ $5 \mid m$ ” fails badly in both directions (33 is a failure with $5 \nmid 33$; 20, 25, 30, 35 are multiples of 5 for which $(1, 1)$ still wins).

The spectral lens supplies the robust statement that survives the failures. In every exceptional case we examined—including $m = 15, 33, 40$ —the *actual* maximizer is the orbit carrying the most near-Nyquist (high-frequency) energy, and it carries strictly more of it than the $(1, 1)$ orbit does. For example at $m = 40$ the maximizer carries 93.7% of its curvature in the near-Nyquist half versus 88.8% for $(1, 1)$. The invariant content of “Fibonacci is special” is thus spectral: among small moduli the classical cycle happens to be the most high-frequency orbit, but for larger moduli other orbits are more jagged still, and curvature tracks jaggedness rather than the Fibonacci seed.

6 What Does Not Hold

We record two negative results, both retiring claims from earlier versions, because closing off false directions is part of the contribution.

The closed form $S_{\max} = m(\pi(m)/2)^2$ is a coincidence. It holds at exactly $m = 9$ (1296) and $m = 27$ (34992) and nowhere else among the moduli tested. It fails already at $m = 3$ (actual 36, predicted 48), and on the next powers of three at $m = 81$ (actual 857304, predicted 944784) and $m = 243$ (actual 21677544, predicted 25509168); it never holds off the powers of three. The appealing reading $1296 = 9F_{12} = 9 \cdot 144$ is numerology specific to $m = 9$. There is no closed form for $S_{\max}(m)$ supported by the data.

The phenomenon does not generalize across recurrences. Replacing the Fibonacci recurrence by $a_{n+1} = p a_n + q a_{n-1}$ destroys “simplest seed maximizes curvature”. For $(p, q) = (1, 2)$ neither $(0, 1)$ nor $(1, 1)$ ever maximizes (over the moduli tested); for the Tribonacci recurrence the natural simple seeds almost never maximize. Only the generator seed $(0, 1)$ under $(p, q) = (2, 1)$ shows a comparable, weaker effect. The clean behaviour is special to Fibonacci.

7 Computational Verification

The script `papers/v4/reproduce.py` in the project repository reproduces every numerical claim in this note, asserting the exact ones: the trichotomy $\{648:24, 972:24, 1296:24\}$

of Theorem 1, the spectral identity of Proposition 1, the unit-class scalings of Remark 3, the half-period split of Section 4.1, the maximality sweep of Section 5, and the failure of the closed form in Section 6. It uses only the Python standard library. The original `verify_curvature.py`, which accompanies Versions 1–3, independently enumerates all 81 seeds modulo 9 and checks the trichotomy. Every numerical claim in this note has thus been verified by at least two independent implementations.

8 Revision History and Errata

This is Version 4. It supersedes Versions 1–3 (doi:10.5281/zenodo.15811767), which remain available as the historical record. The substantive corrections are:

- **The trichotomy is proved by enumeration.** The congruence/“scaling” argument offered as a proof in Versions 1–3 is vacuous (all three curvature values are divisible by 9) and has been replaced by the finite verification, which is the actual proof. The valid residue of that argument—the three values as the three classes of $(\mathbb{Z}/9\mathbb{Z})^\times/\{\pm 1\}$ —is kept as Remark 3.
- **The half-period symmetry claim was wrong.** Version 2 asserted $S = 2 \sum_{n=0}^{11} \kappa_n^2$ from an antisymmetry that the display convention breaks; the true halves are 459 and 837 (Section 4.1). Version 3 already withdrew the explicit formula; this version explains precisely why it fails.
- **The conjecture $S_{\max} = m(\pi(m)/2)^2$ is false** (Section 6); it is a two-point coincidence at $m = 9, 27$.
- **Section 6’s “explanation” is replaced** by the spectral identity (Section 4), which is the genuine reason the maximum is what it is.
- **The novelty claim is softened.** We claim only that we are not aware of prior work applying this spectral/Dirichlet-energy lens to the digital-root orbits of a Pisano period; the underlying circulant diagonalization is classical. A search of OEIS (A030132 and neighbours) and of MathSciNet for “digital root” and “Pisano period” turned up no prior treatment of these second-difference statistics, but we make no exhaustive-priority claim.

9 Open Questions

- Characterize the moduli for which $(1, 1)$ fails to maximize curvature (Section 5); no arithmetic predictor was found, and roughly a third of the failures have a maximizer that is *not* a unit multiple of the Fibonacci orbit.
- Make the spectral heuristic rigorous: prove (or refute) that the curvature maximizer always maximizes the μ_k^2 -weighted near-Nyquist mass.
- Is the $S_{\max} = m(\pi(m)/2)^2$ coincidence at $m = 9, 27$ part of any larger pattern, or genuinely isolated?

Disclosure statement

The author reports no competing interests.

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